

| Assessment | Test | Domain                            | Skill                                                              | Difficulty                                          |
|------------|------|-----------------------------------|--------------------------------------------------------------------|-----------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Medium <div><div></div><div></div><div></div></div> |

Question

15, 14, 18, 17, x

The mean and the median of the five numbers above are equal. Which of the following is NOT a possible value of x ?

Answer

- A. 6
- B. 11
- C. 16
- D. 21

Correct Answer: A

Rationale

Choice A is correct. If x is 6, then the five numbers in the given list are 15, 14, 18, 17, 6. The mean of these five numbers is the sum of all the values divided by the number of values, or  $\frac{15+14+18+17+6}{5} = \frac{70}{5} = 14$ . The median of these five numbers can be found by ordering the numbers from least to greatest and determining the middle value. When ordered from least to greatest, the numbers in the given list are 6, 14, 15, 17, 18, and the middle value is 15. Since the mean is 14 and the median is 15, the mean and median aren't equal when x is 6.

Choices B, C, and D are incorrect. If any of these values is substituted for x, the mean and median of the data set would be equal.